Extracting Conceptual Differences between Translation Pairs Using Multilingual WordNet

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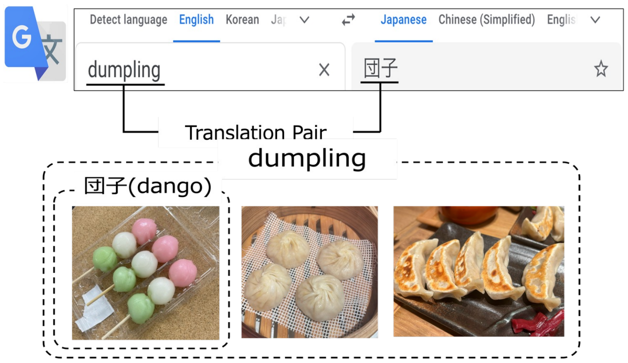
Introduction

How Culture Shapes Language and Meaning

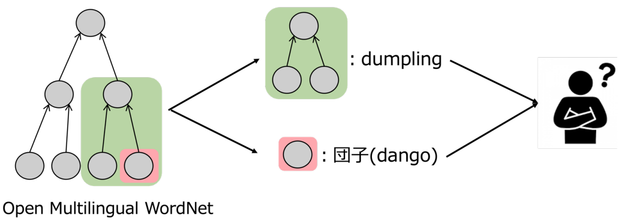
•Languages evolve uniquely, shaped by the cultures they belong to.
•As a result, different languages have distinct conceptual frameworks.
•Even words considered "translations" can convey slightly different meanings.
•These differences in concepts can sometimes lead to misunderstandings in communication.



Method

•**Step 1: Quantify Concepts**
- Use the **Open Multilingual WordNet (OMW)** to map word concepts.
- Define the **conceptual scope** as groups of synsets connected by parent-child or sibling relationships that include the same word.

•**Step 2: Compare Translation Pairs**
- Compare the conceptual scopes of words in translation pairs.
- Identify **conceptual differences** when the scopes differ, highlighting variations in meaning or context.



Conceptual differences were extracted for three target languages: Japanese, Chinese, and Indonesian (Table1).

Table1:Number of Conceptual Differences

Language	Translation Pairs	Conceptual Difference	Ratio
Jp-Zh	236,590	27,005	11.4%
Jp-Id	398,048	60,581	15.2%
Zh-Id	111,450	14,175	12.7%

※Jp:Japanese, Zh:Chinese, Id:Indonesian

Evaluation

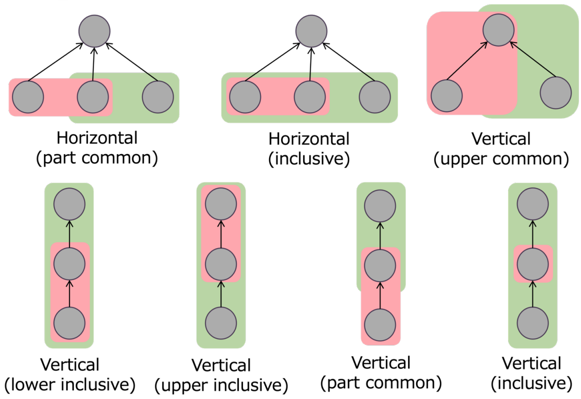
Table2:Assessment results

	Jp-Zh	Jp-Id	Zh-Id
Accuracy	0.81	0.70	0.69
Precision	0.43	0.44	0.32
Recall	0.87	0.80	0.71
F-Score	0.58	0.57	0.44

※Jp:Japanese, Zh:Chinese, Id:Indonesian

Dataset Creation: 100 data samples were randomly selected for each language pair to form evaluation datasets.
Manual Judgment: Conceptual differences between translation pairs were **manually evaluated** to establish ground truth labels.
Performance Evaluation: The ground truth labels were used to assess the accuracy and reliability of our proposed method. Results are summarized in **Table 2**.

Analysis



Sample Selection: 1,000 samples with conceptual differences were randomly selected for evaluation.

Classification: Conceptual differences were grouped into seven types based on word scopes.

Analysis: Chi-square tests analyzed detection accuracy across types.

Results: Detection accuracy was highest for the 'Horizontal (part common)' type in Japanese-Chinese, Japanese-Indonesian, and Chinese-Indonesian pairs.

Acknowledge

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